

| <div> <div>Two Years Normal Operation and Maintenance Spares List</div> <div>As per B957-410-80-42-OM-5310 Rev. A</div> <div>(Validity period of one year and extended for a period of two more years with cost escalation)</div> </div> <div>  </div> |                                                                      |                   |                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-------------------|------------------------------------|
| Sl. No.                                                                                                                                                                                                                                                                                                                                   | Part Description                                                     | Quantity Required | Remarks                            |
|                                                                                                                                                                                                                                                                                                                                           | Based on No. of Installed Units                                      | 1                 |                                    |
| <b>A</b>                                                                                                                                                                                                                                                                                                                                  | <b>Centrifugal pumps (horizontal) one / two stages</b>               |                   | Not applicable to TTL scope        |
| <b>B</b>                                                                                                                                                                                                                                                                                                                                  | <b>Electrical Motor</b>                                              | 1                 | Considered for Lube oil AOP AC-M   |
| 1                                                                                                                                                                                                                                                                                                                                         | Set of bearings (NDE + DE)                                           | 1                 | Not applicable to TTL scope        |
| 2                                                                                                                                                                                                                                                                                                                                         | Terminal box cover with screws                                       | 1                 |                                    |
| 3                                                                                                                                                                                                                                                                                                                                         | Fan                                                                  | 1                 |                                    |
| 4                                                                                                                                                                                                                                                                                                                                         | Terminal block                                                       | 1                 |                                    |
| <b>C</b>                                                                                                                                                                                                                                                                                                                                  | <b>GEAR BOX - GENERAL PURPOSE</b>                                    |                   |                                    |
| 1                                                                                                                                                                                                                                                                                                                                         | Set of Radial Bearings                                               | 1                 |                                    |
| 2                                                                                                                                                                                                                                                                                                                                         | Set of Thrust Bearings                                               | 1                 |                                    |
| 3                                                                                                                                                                                                                                                                                                                                         | Set of Labyrinths, Oil Seals                                         | 1                 |                                    |
| 4                                                                                                                                                                                                                                                                                                                                         | Set of Gaskets, O-rings                                              | 2                 |                                    |
| 5                                                                                                                                                                                                                                                                                                                                         | Complete bare lube oil pump (if integral with gear box.)             | 1                 |                                    |
| 6                                                                                                                                                                                                                                                                                                                                         | Complete Gear box in case                                            | 0                 |                                    |
| <b>D</b>                                                                                                                                                                                                                                                                                                                                  | <b>STEAM TURBINE - GENERAL PURPOSE - API 611</b>                     |                   |                                    |
|                                                                                                                                                                                                                                                                                                                                           | <b>Steam turbine single stage</b>                                    |                   |                                    |
| 1                                                                                                                                                                                                                                                                                                                                         | Complete Rotor dynamically balanced                                  | 1                 |                                    |
| 2                                                                                                                                                                                                                                                                                                                                         | Turbine Shaft                                                        | 0                 |                                    |
| 3                                                                                                                                                                                                                                                                                                                                         | Spare Governor                                                       | 1                 |                                    |
| 4                                                                                                                                                                                                                                                                                                                                         | Set of Radial Bearings (DE + NDE)                                    | 1                 |                                    |
| 5                                                                                                                                                                                                                                                                                                                                         | Set of thrust bearings                                               | 1                 |                                    |
| 6                                                                                                                                                                                                                                                                                                                                         | Set of Bearing Sleeves                                               | 1                 |                                    |
| 7                                                                                                                                                                                                                                                                                                                                         | Set of shaft seal assembly (DE+NDE) including sleeves under seals    | 1                 |                                    |
| 8                                                                                                                                                                                                                                                                                                                                         | Set of gaskets, 'O' rings                                            | 2                 |                                    |
| 9                                                                                                                                                                                                                                                                                                                                         | Set of Carbon Rings                                                  | 8                 | Not applicable                     |
| 10                                                                                                                                                                                                                                                                                                                                        | Springs for carbon Rings                                             | 8                 | Not applicable                     |
| 11                                                                                                                                                                                                                                                                                                                                        | Stoppers for carbon Rings                                            | 8                 | Not applicable                     |
| 12                                                                                                                                                                                                                                                                                                                                        | Set of Gland Housing/Carbon Gland                                    | 1                 | Not applicable                     |
| 13                                                                                                                                                                                                                                                                                                                                        | Set of Fastners for casing, Glands & Bearing housing                 | 1                 |                                    |
| 14                                                                                                                                                                                                                                                                                                                                        | Set of Deflectors with fastners                                      | 1                 | Not applicable                     |
| 15                                                                                                                                                                                                                                                                                                                                        | Set of spares for control valve and ESV including bushing & packings | 1                 | As considered for mandatory spares |
| 16                                                                                                                                                                                                                                                                                                                                        | Governing Valve                                                      | 0                 |                                    |
| 17                                                                                                                                                                                                                                                                                                                                        | Emergency Stop/Trip valve                                            | 0                 |                                    |
| 18                                                                                                                                                                                                                                                                                                                                        | Flexible element of governor Coupling                                | 1                 | Not applicable                     |
| 19                                                                                                                                                                                                                                                                                                                                        | Over speed governor assembly/Trip Body                               | 1                 |                                    |
| 20                                                                                                                                                                                                                                                                                                                                        | Completed Coupling of governor                                       | 1                 | Not applicable                     |
| 21                                                                                                                                                                                                                                                                                                                                        | Labyrinth seal assembly                                              | 1                 | Not applicable                     |
| 22                                                                                                                                                                                                                                                                                                                                        | Full coupling assembly                                               | 1                 |                                    |
| 23                                                                                                                                                                                                                                                                                                                                        | Flexible element assembly for coupling with fasteners                | 1                 |                                    |
| 24                                                                                                                                                                                                                                                                                                                                        | Lube of filters                                                      | 2                 |                                    |
| 25                                                                                                                                                                                                                                                                                                                                        | Main oil pump                                                        | 1                 |                                    |
| 26                                                                                                                                                                                                                                                                                                                                        | Mechanical seal assy                                                 | 2                 |                                    |

Note - The above Tender spare list is same for the Both tag of turbines 410-PT-00-108 & 411-PT-00-109 .